

Master's Mathematics
2026-27*

**Transitional provisions apply; discuss with Study Advisor and/or see TER 2026-27, Article 11.2.*

COMPULSORY components		54 or 60 EC	
Course Name	Code	EC	
p1+2 Current Issues in Mathematics	XM_0202	6	
Master Project Mathematics	X_400355	36	
Contrained choice in seminars: choose 2 or 3 throughout studies*			
p1+2 Data Assimilation**	XM_0203	6	
p1+2 Sequential Monte Carlo Methods	XM_0182	6	
p4+5 Geometry of Statistical Models	XM_0204	6	
p4+5 Covering Spaces in Topology and Algebra	XM_0205	6	
<i>*offer may vary from year to year</i>			
<i>**not in combination with XM_0183 Uncertainty Quantification offered in 2025-26</i>			

Mathematics and Data study path

STRONG RECOMMENDATIONS for electives		>=30 EC	
Course Name	Code	EC	
p1 Mathematical Foundations of Machine Learning	XM_0189	6	
p1+2 Statistical Models	X_400418	6	
p2 Mathematical Optimization	XM_0051	6	
Choose at least 2 from:			
p1+2 Applied Stochastic Modelling	X_400392	6	
p2 Dynamic Programming and Reinforcement Learning	XM_0093	6	
p4 Large Language Models from scratch	XM_0188	6	
p4+5 Optimization of Business Processes	X_400422	6	
p4+5 Statistical Learning with High-Dimensional Data	XM_0206	6	
fall Asymptotic Statistics	Mastermath	8	
fall Measure Theoretic Probability	Mastermath	8	
spring Stochastic Processes	Mastermath	8	

FURTHER RECOMMENDATIONS for electives		<=36 or 30 EC	
Course Name	Code	EC	
Match department's research profile, will prepare you particularly well for your Master Project:			
p2 Performance of Networked Systems	X_405105	6	
p1+2 Stochastic Processes for Finance	X_400352	6	
fall Continous Optimization	Mastermath	6	
fall Forensic Probability and Statistics	Mastermath	8	
fall Time Series	Mastermath	8	
fall High-dimensional Probability	Mastermath	8	
spring Machine Learning Theory	Mastermath	8	
spring Scheduling Theory and Algorithms	Mastermath	6	
spring Queueing Theory	Mastermath	6	
Extra:			
p4+5 Applied Analysis: Financial Mathematics (MSc)	XM_0140	6	
fall Discrete Optimization	Mastermath	6	
spring Causality	Mastermath	8	

Mathematics and Computation study path

STRONG RECOMMENDATIONS for electives		>=22 EC	
Course Name	Code	EC	
p1 Mathematical Foundations of Machine Learning	XM_0189	6	
Choose at least 2 from:			
fall Algebraic Topology I	Mastermath	8	
fall Dynamical Systems	Mastermath	8	
fall Partial Differential Equations	Mastermath	8	
Mathematical Biology*	Mastermath	8	
spring Mathematical Neuroscience	Mastermath	8	
spring Elliptic Curves	Mastermath	8	
spring Formal Methods in Mathematics	Mastermath	8	
Topological Data Analysis**	Mastermath	8	
<i>*alternates with Mathematical Neuroscience, not offered in 2026-27</i>			
<i>**offered every other year, not in 2026-27</i>			

FURTHER RECOMMENDATIONS for electives, profile in Biomedical Mathematics		<=44 or 38 EC	
Course Name	Code	EC	
Match department's research profile, will prepare you particularly well for your Master Project:			
p4+5 Statistical Learning with High-Dimensional Data	XM_0206	6	
fall Measure Theoretic Probability	Mastermath	8	
fall Numerical Methods for PDEs	Mastermath	8	
spring Machine Learning Theory	Mastermath	8	
spring Stochastic Processes	Mastermath	8	
Extra:			
fall Functional Analysis	Mastermath	8	
spring Inverse Problems in Imaging	Mastermath	6	
p1 From Molecule to Mind	AM_1275	6	
p1 Fundamentals of Bioinformatics	X_405052	6	
p1 Introduction to Systems Biology	X_428565	6	
p2 Algorithms in Sequence Analysis	X_405050	6	
p2 Mechanics and Thermodynamics in the Cell	X_422589	6	
p4 Basic Models of Biological Networks	X_418154	6	
p4 Brain Imaging (for AI)	P_MBRIMA_AI	6	
p4 Structural Bioinformatics	X_405019	6	
p5 Bioinformatics for Translational Medicine	X_405092	6	
p5 Tracer Kinetic Modelling	XM_0014	6	

FURTHER RECOMMENDATIONS for electives, profile in Algebra, Geometry and Number Theoy		<=44 or 38 EC	
Course Name	Code	EC	
Match department's research profile, will prepare you particularly well for your Master Project:			
fall Algebraic Number Theory	Mastermath	8	
fall Analytic Number Theory	Mastermath	8	
fall Modern Cryptography	Mastermath	8	
fall Algebraic Geometry I	Mastermath	8	
fall Advanced Algebraic Geometry	Mastermath	8	
fall Differential Geometry	Mastermath	8	
spring Algebraic Geometry II	Mastermath	8	
spring Algebraic Topology II	Mastermath	8	
spring Symplectic Geometry	Mastermath	8	
Extra:			
fall Commutative Algebra	Mastermath	8	
spring Coding Theory	Mastermath	8	
spring Selected Areas in Cryptology	Mastermath	8	

FURTHER RECOMMENDATIONS for electives, profile in Analysis and Dynamical Systems		<=44 or 38 EC	
Course Name	Code	EC	
Match department's research profile, will prepare you particularly well for your Master Project:			
fall Numerical Methods for PDEs	Mastermath	8	
fall Differential Geometry	Mastermath	8	
spring Numerical Methods for SDEs	Mastermath	8	
spring Non-Inear PDEs	Mastermath	8	
spring Calculus of Variations	Mastermath	8	
spring Interacting Particle Systems	Mastermath	8	
spring Symplectic Geometry	Mastermath	8	
Extra:			
p4+5 Applied Analysis: Financial Mathematics (MSc)	XM_0140	6	
fall Functional Analysis	Mastermath	8	
spring Inverse Problems in Imaging	Mastermath	6	

FULL LIST of ELECTIVES		
Course Name	Code	EC
<i>- all local courses from above (code other than 'Mastermath')</i>		
<i>- an additional local course:</i>		
p4 Entrepreneurship in Analytics and AI	XM_0090	6
<i>- all Mastermath courses</i>		
<i>- up to 2 level-300 courses from BSc Mathematics: advance permission from the Exam Board required</i>		